Batting and Pitching Metric's Evolution in College Baseball: A Time Series and Bayesian Analysis

Mateo Biggs & Jeremy Marino

Northeastern University

Introduction / Abstract

Baseball, like many other sports at the turn of the twentieth century, has experienced a renaissance in statistical analysis. Foundational studies, most famously Hakes and Sauer's Moneyball Hypothesis [1], laid the groundwork for baseball (and all other sports) analytics, discovering undervalued metrics, particularly on-base percentage, that had strong explanatory value in relation to winning baseball games. Since this phenomenon, baseball has undergone another analytic revolution focused on home runs, strikeouts, and the expected value of a given duel. At the professional level, teams have found that rather than getting players on-base at a high rate, swinging with the intent of hitting a home run provides a greater expected run count [2]. As a result, batters are incentivized to swing hard at the cost of their batting average, in turn increasing the rate of strikeouts from a pitcher.

While this trend has become more prevalent in the MLB, it is unclear if its effects have cascaded to lower levels of baseball. Our objective is to identify how these trends have affected the 3 divisional levels of collegiate baseball, if their effects have been uniform across divisions, and how these values may be projected to evolve moving forward.

Data & Exploration

All team-level season statistics were scraped from `stats.ncaa.org`, covering Divisions 1, 2, and 3. Each team-season record includes standard batting metrics (OBP, SLG, runs scored, batting average, home runs per game) and pitching metrics (ERA, WHIP, K/9, runs allowed). After cleaning, the dataset contains over 12,261 team-season observations across approximately 130 conferences and 1000 unique team seasons (2011-2025 alone).

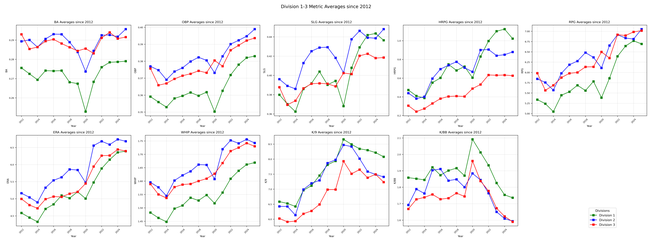

Fig. 1: Division 1–3 Target Metric Trends

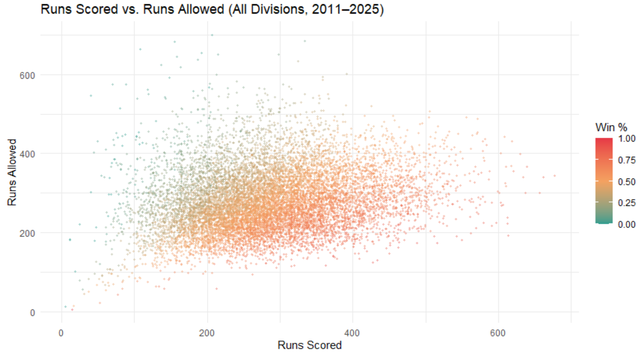

Fig. 2: Scatterplot of runs scored vs. runs allowed colored by win percentage

Methodology

Time Series: Autoregressive Model

The AR model uses a dummy-based division model, which allows us to get the metric averages of individual conferences and weight them based on how over or under representative they are in the division (for example, a 3.50 average ERA in the SEC may be different relative to a 3.50 ERA in the Big 12). Our forecasting model takes the average of our coefficients found in our dummy values and predicts the trajectory of the given metric with a range of uncertainty. This allows cautious predicting moving forward based on the data we have, and can give some insight into how the relationship between 2 metrics could evolve over time.

$$y_t = \alpha + \beta_1 y_{t-1} + \sum_{j=1}^{K} \gamma_j \cdot \text{Conf}_j + \epsilon_t$$

Bayesian: Hierarchical Linear Regression

The Bayesian model estimates the effect of run production and run prevention on win percentage, allowing coefficients to vary by season with conference-level random effects. All parameters are estimated via a hand-coded Metropolis-Hastings sampler (10,000 iterations, 5,000 burn-in) rather than a package.

$$\text{WPCT} = \alpha + \beta^{\text{batting}} \cdot R_{\text{batting}} + \beta^{\text{pitching}} \cdot R_{\text{pitching}} + \gamma + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

$$\alpha \sim \mathcal{N}(0.5,\, 0.2^2), \quad \beta^{\text{batting}},\, \beta^{\text{pitching}} \sim \mathcal{N}(0,\, 0.5^2), \quad \gamma_c \sim \mathcal{N}(0,\, 0.1^2)$$

$$p(\theta \mid y) \propto p(y \mid \theta)\, p(\theta)$$

Time Series Results

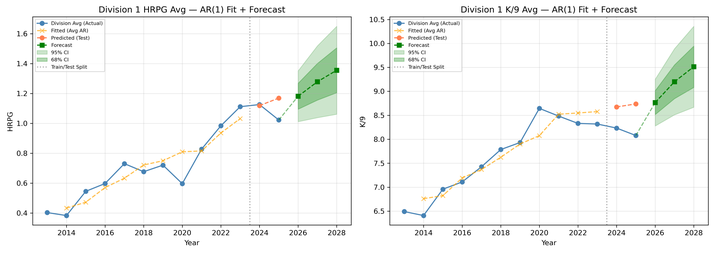

Fig. 3: AR Model Forecast for Division 1 HRPG & K/9 (2012-2025)

Key Takeaways:

- **How have pitching and batting metrics evolved in at the collegiate level?**
  While it was the case that home run rates are rising, we have a wildly uncertain state for pitcher's strikeout rates. However, there is stronger evidence that batters are scoring runs more frequently due to the forecasted increase in ERA and WHIP from pitchers. The future movement of these metrics seem to imply that batters are getting better across the board (in hitting home runs and getting on base) without sacrificing their batting average substantially.
- **Are these changes uniform amongst all divisions?**
  Generally metric trends followed similar patterns between divisions, with division 3 having the most metric variance.
- **Can we effectively project how these metrics will change moving forward?**
  Some yes, particularly the metrics with linear trends such as WHIP and ERA. Some of the more variant trends (K/9 in particular) did not fit our AR model as successfully.

Bayesian Model Results

Our Bayesian hierarchical model estimates season-varying coefficients for runs scored (batting) and runs allowed (pitching) on win percentage, with conference-level random effects estimated via a hand-coded Metropolis-Hastings sampler across all three NCAA divisions (2011–2025, excluding 2020).

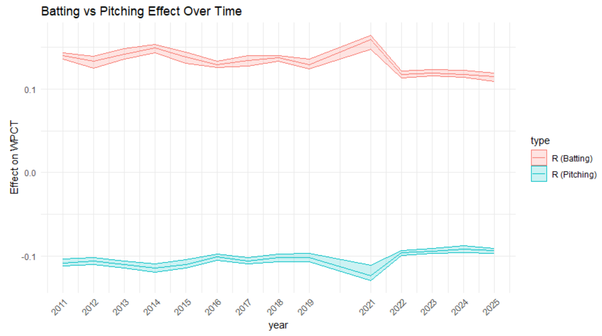

Fig. 4: Season-Varying Batting and Pitching Effects on Win Percentage (2011–2025)

Key Takeaways:

- **Runs (batting) has a stronger direct effect on winning.**
  The R (Batting) coefficient (0.12–0.16) consistently exceeds the magnitude of R (Pitching) (-0.09 to -0.12), suggesting that scoring runs has greater explanatory power for win percentage than preventing them across all three divisions.
- **Run prevention's stable while batting declines.**
  The R (Pitching) effect remains nearly flat across 14 seasons with tight credible intervals, while 2021 onward, the R (Batting) coefficient drops from 0.16 to 0.12 with widening uncertainty.

Discussion / Future Work

- **Small Sample Size**
  Models only have 14 years of data to train on. Future data will help flesh out trends and validate our results.
- **Metric Scope**
  Finding relationships between more metrics could be useful to widen our scope and understanding of these changes at all 3 levels.
- **Moving Average Model**
  Continue development moving forward to see what addition parameters/frameworks we can use to fit our less linear trends effectively.

References

[1] Jahn K. Hakes and Raymond D. Sauer. "An Economic Evaluation of the Moneyball Hypothesis". In: *Journal of Economic Perspectives* 20.3 (2006), pp. 173–186. DOI: 10.1257/jep.20.3.173.

[2] Tom M. Tango, Mitchel G. Lichtman, and Andrew E. Dolphin. *The Book: Playing the Percentages in Baseball*. Washington, D.C.: Potomac Books, 2007.